

Experiment: 09

Aim: To implement a Private Ethereum Blockchain using Geth.

Theory:

Ethereum is an open-source, decentralized blockchain platform that enables developers to build and deploy smart contracts and decentralized applications (DApps). Unlike public blockchains such as the Ethereum mainnet, a Private Ethereum Blockchain is restricted to specific participants, making it ideal for development, testing, and enterprise applications where confidentiality, control, and performance are essential.

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through JSON-RPC, command-line, or JavaScript consoles.

Setting up a **Private Ethereum Network** involves creating a customized blockchain environment isolated from the public Ethereum network. It enables developers to test smart contracts, consensus mechanisms, and network behavior without incurring gas costs or depending on public validators.

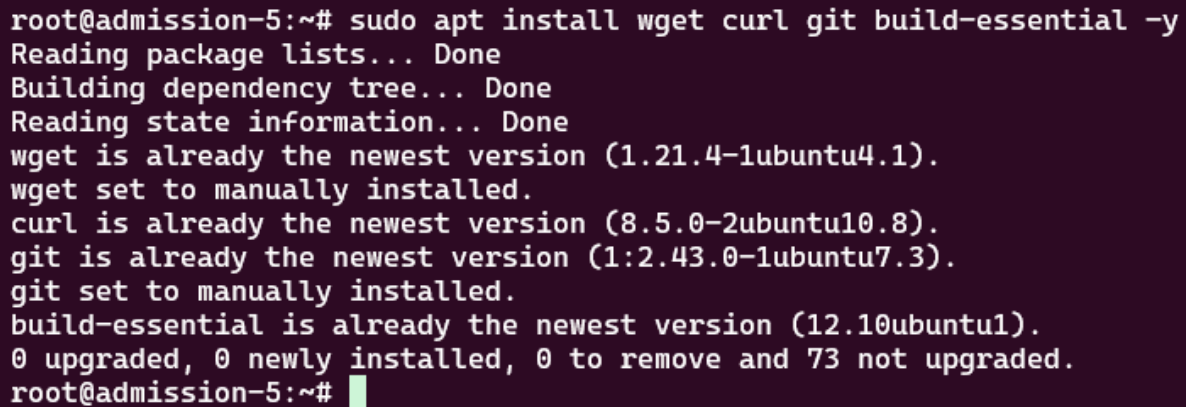
Steps Involved:

1. **Choosing a Network ID:**
2. Each private blockchain must have a unique network ID to differentiate it from public or other private networks.
3. **Choosing a Consensus Algorithm:**
4. Ethereum supports different consensus mechanisms such as **Proof of Work (PoW)** and **Proof of Authority (PoA)**. In private networks, **Clique (PoA)** is often used for faster block confirmation.
5. **Creating a Genesis Block:**
6. The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.
7. **Initializing the Geth Database:**
8. The `geth init` command initializes the blockchain using the `genesis.json` file, preparing the data directory for node operation.
9. **Setting up Networking:**
10. Nodes communicate using **enodes** (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.
11. **Running the Member Nodes:**
12. Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.
13. **Running a Signer (in Clique):**
14. In the **Clique consensus**, designated **signers** validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

Implementation:

1. First install all the necessary dependencies like curl, git, build-essential.

sudo apt install wget curl git build-essential -y

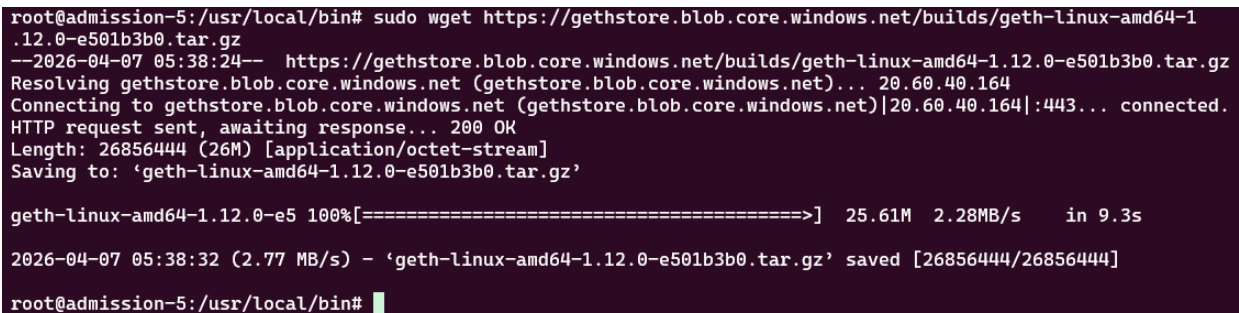


```
root@admission-5:~# sudo apt install wget curl git build-essential -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.8).
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 73 not upgraded.
root@admission-5:~#
```

2. Install geth using wget

cd /usr/local/bin && sudo wget

<https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e501b3b0.tar.gz>



```
root@admission-5:/usr/local/bin# sudo wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e501b3b0.tar.gz
--2026-04-07 05:38:24-- https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e501b3b0.tar.gz
Resolving gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)... 20.60.40.164
Connecting to gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)|20.60.40.164|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 26856444 (26M) [application/octet-stream]
Saving to: 'geth-linux-amd64-1.12.0-e501b3b0.tar.gz'

geth-linux-amd64-1.12.0-e5 100%[=====] 25.61M 2.28MB/s in 9.3s
2026-04-07 05:38:32 (2.77 MB/s) - 'geth-linux-amd64-1.12.0-e501b3b0.tar.gz' saved [26856444/26856444]

root@admission-5:/usr/local/bin#
```

3. Extract the tar file and move the geth into /usr/local/bin/geth path & check the geth version

sudo tar -xvf geth-linux-amd64-1.12.0-e501b3b0.tar.gz sudo mv geth-linux-amd64-1.12.0-e501b3b0/geth /usr/local/bin/geth geth version

```

root@admission-5:/usr/local/bin# sudo tar -xvf geth-linux-amd64-1.12.0-e501b3b0.tar.gz
sudo mv geth-linux-amd64-1.12.0-e501b3b0/geth /usr/local/bin/geth
geth version
geth-linux-amd64-1.12.0-e501b3b0/
geth-linux-amd64-1.12.0-e501b3b0/COPYING
geth-linux-amd64-1.12.0-e501b3b0/geth
Geth
Version: 1.12.0-stable
Git Commit: e501b3b05db8e169f67dc78b7b59bc352b3c638d
Git Commit Date: 20230525
Architecture: amd64
Go Version: go1.20.3
Operating System: linux
GOPATH=
GOROOT=
root@admission-5:/usr/local/bin# █

```

4. Create a folder named `private_ethereum_setup` and create another 2 directories naming `node1` and `node2` individually inside `private_ethereum_setup` directory.

`mkdir private_ethereum_setup cd private_ethereum_setup && mkdir node1 node2`

```

root@admission-5:/usr/local/bin# mkdir private_ethereum_setup
root@admission-5:/usr/local/bin# cd private_ethereum_setup && mkdir node1 node2
root@admission-5:/usr/local/bin/private_ethereum_setup# ls
node1 node2
root@admission-5:/usr/local/bin/private_ethereum_setup# █

```

5. Create account for `node1`

`geth --datadir "node1" account new`

```

root@admission-5:/usr/local/bin/private_ethereum_setup# geth --datadir "node1" account new
INFO [04-07|05:42:29.002] Maximum peer count      ETH=50 LES=0 total=50
INFO [04-07|05:42:29.003] Smartcard socket not found, disabling  err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:  0xB39388323dCD82eeb28F73B3448424c008FE7150
Path of the secret key file: node1/keystore/UTC--2026-04-07T05-42-42.824513266Z--b39388323dcd82eeb28f73b3448424c008fe7150

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
root@admission-5:/usr/local/bin/private_ethereum_setup# █

```

6. Do the same for `node2`

`geth --datadir "node2" account new`

```
root@admission-5:/usr/local/bin/private_ethereum_setup# geth --datadir "node2" account new
INFO [04-07|05:43:28.087] Maximum peer count          ETH=50 LES=0 total=50
INFO [04-07|05:43:28.087] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key: 0x126F82349979c6b3d2a52eE032b7ea11880D2dC9
Path of the secret key file: node2/keystore/UTC--2026-04-07T05-43-35.386933441Z--126f82349979c6b3d2a52ee032b7ea11880d2dc9

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!

root@admission-5:/usr/local/bin/private_ethereum_setup#
```

7. Create a file called `genesis.json` in the same directory and copy paste these contents

[illegible]

```
"balance": "100000000000000000000"
```

8. Cat genesis.json

```
root@admission-5:/usr/local/bin/private_ethereum_setup# cat genesis.json
{
  "config": {
    "chainId": 12345,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "istanbulBlock": 0,
    "berlinBlock": 0,
    "londonBlock": 0,
    "clique": {
      "period": 5,
      "epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "8000000",
  "extradata":
"0x0000000000000000000000000000000000000000000000000000000000000000
00000000c99cf64ef49425bf0e814c9d51364281a61747d200000000000000
0000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000
00000",
  "alloc": {
    "0xB39388323dCD82eeb28F73B3448424c008FE7150": {
      "balance": "1000000000000000000000"
    },
    "0x126F82349979c6b3d2a52eE032B7ea11880D2dC9": {
      "balance": "1000000000000000000000"
    }
  }
}
```

9. Initializing geth for node1 using the genesis.json file

```
geth init --datadir node1 genesis.json
```

```

root@admission-5:/usr/local/bin/private_ethereum_setup# geth init --datadir node1 genesis.json
INFO [04-07|05:58:25.537] Maximum peer count          ETH=50 LES=0 total=50
INFO [04-07|05:58:25.538] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-07|05:58:25.543] Set global gas cap          cap=50,000,000
INFO [04-07|05:58:25.544] Initializing the KZG library backend=gokzg
INFO [04-07|05:58:25.594] Defaulting to pebble as the backing database
INFO [04-07|05:58:25.596] Allocated cache and file handles database=/usr/local/bin/private_ethereum_setup/node1/geth/chaindata
a cache=16.00MiB handles=16
INFO [04-07|05:58:25.618] Opened ancient database      database=/usr/local/bin/private_ethereum_setup/node1/geth/chaindata
a/ancient/chain readonly=false
INFO [04-07|05:58:25.619] Writing custom genesis block
INFO [04-07|05:58:25.619] Persisted trie from memory database nodes=3 size=411.00B time="9.642µs" gcnodes=0 gcsiz=0.00B gctime=
0s livenodes=0 livesize=0.00B
INFO [04-07|05:58:25.623] Successfully wrote genesis state database=chaindata hash=8c5dd7..24caed
INFO [04-07|05:58:25.623] Defaulting to pebble as the backing database
INFO [04-07|05:58:25.623] Allocated cache and file handles database=/usr/local/bin/private_ethereum_setup/node1/geth/lightcha
indata cache=16.00MiB handles=16
INFO [04-07|05:58:25.640] Opened ancient database      database=/usr/local/bin/private_ethereum_setup/node1/geth/lightcha
indata/ancient/chain readonly=false
INFO [04-07|05:58:25.641] Writing custom genesis block
INFO [04-07|05:58:25.641] Persisted trie from memory database nodes=3 size=411.00B time="10.121µs" gcnodes=0 gcsiz=0.00B gctime
=0s livenodes=0 livesize=0.00B
INFO [04-07|05:58:25.644] Successfully wrote genesis state database=lightchaindata hash=8c5dd7..24caed
root@admission-5:/usr/local/bin/private_ethereum_setup#

```

10. Do the same for node2

geth init --datadir node2 genesis.json

```

root@admission-5:/usr/local/bin/private_ethereum_setup# geth init --datadir node2 genesis.json
INFO [04-07|05:59:40.725] Maximum peer count          ETH=50 LES=0 total=50
INFO [04-07|05:59:40.726] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-07|05:59:40.729] Set global gas cap          cap=50,000,000
INFO [04-07|05:59:40.729] Initializing the KZG library backend=gokzg
INFO [04-07|05:59:40.760] Defaulting to pebble as the backing database
INFO [04-07|05:59:40.760] Allocated cache and file handles database=/usr/local/bin/private_ethereum_setup/node2/geth/chaindata
a cache=16.00MiB handles=16
INFO [04-07|05:59:40.780] Opened ancient database      database=/usr/local/bin/private_ethereum_setup/node2/geth/chaindata
a/ancient/chain readonly=false
INFO [04-07|05:59:40.780] Writing custom genesis block
INFO [04-07|05:59:40.780] Persisted trie from memory database nodes=3 size=411.00B time="10.186µs" gcnodes=0 gcsiz=0.00B gctime
=0s livenodes=0 livesize=0.00B
INFO [04-07|05:59:40.784] Successfully wrote genesis state database=chaindata hash=8c5dd7..24caed
INFO [04-07|05:59:40.784] Defaulting to pebble as the backing database
INFO [04-07|05:59:40.784] Allocated cache and file handles database=/usr/local/bin/private_ethereum_setup/node2/geth/lightcha
indata cache=16.00MiB handles=16
INFO [04-07|05:59:40.805] Opened ancient database      database=/usr/local/bin/private_ethereum_setup/node2/geth/lightcha
indata/ancient/chain readonly=false
INFO [04-07|05:59:40.805] Writing custom genesis block
INFO [04-07|05:59:40.805] Persisted trie from memory database nodes=3 size=411.00B time="10.09µs" gcnodes=0 gcsiz=0.00B gctime
=0s livenodes=0 livesize=0.00B
INFO [04-07|05:59:40.808] Successfully wrote genesis state database=lightchaindata hash=8c5dd7..24caed
root@admission-5:/usr/local/bin/private_ethereum_setup#

```

11. Now we will install bootnode for configuring both the nodes.

sudo add-apt-repository -y ppa:ethereum/ethereum

sudo apt-get update

sudo apt-get install bootnode

```

root@admission-5:/usr/local/bin/private_ethereum_setup# sudo add-apt-repository -y ppa:ethereum/ethereum
sudo apt-get update
sudo apt-get install bootnode
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu/
Suites: noble
Components: main
'
More info: https://launchpad.net/~ethereum/+archive/ubuntu/ethereum
Adding repository.
Hit:1 https://deb.nodesource.com/node_18.x nodistro InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:6 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble InRelease [18.1 kB]
Get:7 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main amd64 Packages [2576 B]
Get:8 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main Translation-en [772 B]
Fetched 21.4 kB in 1s (18.3 kB/s)
Reading package lists... Done
Hit:1 https://deb.nodesource.com/node_18.x nodistro InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  bootnode
0 upgraded, 1 newly installed, 0 to remove and 73 not upgraded.

```

12. Create a key for the bootnode and save it to boot.key in the folder
bootnode -genkey boot.key

```

root@admission-5:/usr/local/bin/private_ethereum_setup# bootnode -genkey boot.key
root@admission-5:/usr/local/bin/private_ethereum_setup# ls
boot.key  genesis.json  node1  node2
root@admission-5:/usr/local/bin/private_ethereum_setup# █

```

13. Copy this command and copy the enode://.... This will be required later
bootnode -nodekey boot.key -verbosity 9 -addr :30305

```

root@admission-5:/usr/local/bin/private_ethereum_setup# bootnode -nodekey boot.key -verbosity 9 -addr :30305
enode://7fff7a90d47bc73762802a4f924fdb403af4bfff08ded3c3b0b5c2efeb9f76553859f6edf9abdf20619c04c9408fc7ed606e097a26d4519d50f5afcc063e02120@127.0.0.1:30305
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [04-07|06:03:30.800] New local node record          seq=1,775,541,810,798 id=f572303c2cd66d79 ip="invalid IP" udp=0 tc
p=0

```

14. Now on another terminal we will start mining node1

```

geth --datadir node1 \
--port 30306 \
--bootnodes
"enode://7fff7a90d47bc73762802a4f924fdb403af4bfff08ded3c3b0b5c2efeb9f76553859f6edf9abdf20619c04c9408fc7ed606e097a26d4519d50f5afcc063e02120@127.0.0.1:30305" \
--networkid 123454321 \
--unlock 0xB39388323dCD82eeb28F73B3448424c008FE7150 \
--password node1/password.txt \

```



```

root@admission-5: /usr/local/bin/private_ethereum_setup# rm -rf /usr/local/bin/private_ethereum_setup/node1/gets
root@admission-5: /usr/local/bin/private_ethereum_setup# geth --datadir node1 --port 30306 --bootnodes "enode://7fff9a90d47bc73762802a4f924fcd403af4bfff08ded3c3b0b5c2efeb9f76553859f6edf9abdf20619c04c9408c7ed606e997a26d4519d50f5afcc063e02120@127.0.0.1:30305" --networkid 123454321 --unlock 0xb39388323dCD82eeb28f73B3448424c008FE7150 --password node1/password.txt --authrpc.port 8551 --miner.etherbase 0xb39388323dCD82eeb28f73B3448424c008FE7150 --mine
INFO [04-07]06:38:47.164] Maximum peer count ETH=50 LES=0 total=50
INFO [04-07]06:38:47.166] Startcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-07]06:38:47.168] Set global gas cap cap=50,000,000
INFO [04-07]06:38:47.169] Initializing the KZG library backend=gokzg
INFO [04-07]06:38:47.199] Allocated trie memory caches clean=154.00MiB dirty=256.00MiB
INFO [04-07]06:38:47.199] Defaulting to pebble as the backing database
INFO [04-07]06:38:47.199] Allocated cache and file handles database=/usr/local/bin/private_ethereum_setup/node1/gets/chaindata
a cache=512.00MiB handles=524.288
INFO [04-07]06:38:47.225] Opened ancient database database=/usr/local/bin/private_ethereum_setup/node1/gets/chaindata
a/ancient/chain readonly=false
INFO [04-07]06:38:47.225] Initialising Ethereum protocol network=123,454,321 dbversion=<nil>
INFO [04-07]06:38:47.226] Writing default main-net genesis block
INFO [04-07]06:38:47.461] Persisted trie from memory database nodes=12356 size=1.79MiB time=48.92926ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 liveness=0.00B
INFO [04-07]06:38:47.476] -----
INFO [04-07]06:38:47.476] Chain ID: 1 (mainnet)
INFO [04-07]06:38:47.476] Consensus: Beacon (proof-of-stake), merged from Ethash (proof-of-work)
INFO [04-07]06:38:47.476]
INFO [04-07]06:38:47.476] Pre-Merge hard forks (block based):
INFO [04-07]06:38:47.476] - Homestead: #1150000 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/homestead.md)
INFO [04-07]06:38:47.476] - DAO Fork: #1920000 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/dao-fork.md)
INFO [04-07]06:38:47.476] - Tangerine Whistle (EIP 150): #2463000 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/tangerine-whistle.md)

root@admission-5: ~
X root@admission-5: /usr/local/ X root@admission-5: /usr/local/ X + v - - -
WARN [04-07]06:53:41.791] Unclean shutdown detected
WARN [04-07]06:53:41.791] Engine API enabled
INFO [04-07]06:53:41.791] Starting peer-to-peer node
INFO [04-07]06:53:41.799] Generated state snapshot
INFO [04-07]06:53:41.806] New local node record
cp=30306
INFO [04-07]06:53:41.806] Started P2P networking
09a7d87a8d4c6ccab873cdb8e40fb59e0d5b9d3063d1a2e7b1c0042de2863090ad672c3b45a0127.0.0.1:30306
INFO [04-07]06:53:41.807] IPC endpoint opened
INFO [04-07]06:53:41.807] Loaded JWT secret file
c32=0x21e9937f
INFO [04-07]06:53:41.808] WebSocket enabled
INFO [04-07]06:53:41.808] HTTP server started
calhost
INFO [04-07]06:53:42.426] Unlocked account
INFO [04-07]06:53:42.426] Transaction pool price threshold updated
INFO [04-07]06:53:42.426] Transaction pool price threshold updated
INFO [04-07]06:53:42.427] Commit new sealing work
ed="787.008µs"
panic: ethash (pow) sealing not supported any more

goroutine 46 [running]:
github.com/ethereum/go-ethereum/consensus/ethash.(*Ethash).Seal(0xc0002f6520?, {0x41aea6?, 0xc00044c2c0?}, 0xc00044c3c8?, 0xc00044c2c0?, 0x41aea6bcd6d815a31?)
github.com/ethereum/go-ethereum/consensus/ethash/ethash.go:84 +0x27
github.com/ethereum/go-ethereum/consensus/beacon.(*Beacon).Seal(0xc000496a80, {0x2066f20, 0xc000581800}, 0xc0002c6640, 0x9daddf69ca147c60?, 0x9e5c859464211001?)
github.com/ethereum/go-ethereum/consensus/beacon/consensus.go:387 +0x9e
github.com/ethereum/go-ethereum/miner.(*worker).taskLoop(0xc0001ce280)
github.com/ethereum/go-ethereum/miner/worker.go:706 +0x2fc
created by github.com/ethereum/go-ethereum/miner.newWorker
github.com/ethereum/go-ethereum/miner/worker.go:329 +0xa8d
root@admission-5: /usr/local/bin/private_ethereum_setup#

booted=2026-04-07T06:38:47+0000 age=14m54s
protocol=eth
instance=Geth/v1.12.0-stable-e501b3b0/linux-amd64/go1.20.3
accounts=8893 slots=0 storage=409.64KiB dangling=0 elapsed=9.332ms
seq=1,775,543,927,501 id=5fffab40659cdeaba ip=127.0.0.1 udp=30306 t
self=enode://45d557b46f2495e3c7bdf4d401c3e3025b83bbf5e8f9b9badb57a672c3b45a0127.0.0.1:30306
url=/usr/local/bin/private_ethereum_setup/node1/gets.ipc
path=/usr/local/bin/private_ethereum_setup/node1/gets/jwtsecret cr
url=ws://127.0.0.1:8551
endpoint=127.0.0.1:8551 auth=true prefix= cors=localhost vhosts=lo
address=0xb39388323dCD82eeb28f73B3448424c008FE7150
price=0
price=1,000,000,000
number=1 sealhash=0dc167.0b853c uncles=0 txs=0 gas=0 fees=0 elaps

```

15. Now do same for node2


```

pranav@DESKTOP-OSHSLJ1: ~/private_ethereum_setup$ geth --datadir node2 \
--port 30307 \
--networkid 12345 \
--bootnodes "enode://6510fcbf2bac27c639de2d15c6265f8135751d09041a24880666ef4a4a206a0bf0673458a5dcb1146bda9aa54d511cbcabfd26c807257191ec1c77b46472@127.0.0.1:30305" \
--http --http.port 8546 --http.api "eth,net,web3" \
--authrpc.port 8552
INFO [03-31|14:18:32.384] Maximum peer count                      ETH=50 LES=0 total=50
INFO [03-31|14:18:32.385] Smartcard socket not found, disabling   err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [03-31|14:18:32.387] Set global gas cap                       cap=50,000,000
INFO [03-31|14:18:32.387] Initializing the KZG library             backend=gokzg
INFO [03-31|14:18:32.433] Allocated trie memory caches            clean=154.00MiB dirty=256.00MiB
INFO [03-31|14:18:32.433] Using pebble as the backing database
INFO [03-31|14:18:32.433] Allocated cache and file handles        database=/home/pranav/private_ethereum_setup/node2/geth/chaindata cache=512.00MiB handles=524,288
INFO [03-31|14:18:32.479] Opened ancient database                 database=/home/pranav/private_ethereum_setup/node2/geth/chaindata/ancient/chain_readonly=false
INFO [03-31|14:18:32.479] Initialising Ethereum protocol          network=12345 dbversion=nil>
INFO [03-31|14:18:32.480] -----
INFO [03-31|14:18:32.480] Chain ID: 12345 (unknown)
INFO [03-31|14:18:32.480] Consensus: Clique (proof-of-authority)
INFO [03-31|14:18:32.480]
INFO [03-31|14:18:32.480] Pre-Merge hard forks (block based):
INFO [03-31|14:18:32.480]   - Homestead: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/homestead.md)
INFO [03-31|14:18:32.480]   - Tangerine Whistle (EIP 150): #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/tangerine-whistle.md)
INFO [03-31|14:18:32.480]   - Spurious Dragon/1 (EIP 155): #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/spurious-dragon.md)
INFO [03-31|14:18:32.480]   - Spurious Dragon/2 (EIP 158): #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/spurious-dragon.md)
INFO [03-31|14:18:32.480]   - Byzantium: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/byzantium.md)
INFO [03-31|14:18:32.480]   - Constantinople: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/constantinople.md)
INFO [03-31|14:18:32.513] Started P2P networking
INFO [03-31|14:18:32.516] IPC endpoint opened
INFO [03-31|14:18:32.517] Generated JWT secret
INFO [03-31|14:18:32.517] HTTP server started
INFO [03-31|14:18:32.518] WebSocket enabled
INFO [03-31|14:18:32.518] HTTP server started
INFO [03-31|14:18:43.349] Block synchronisation started
INFO [03-31|14:18:43.365] Syncing: state download in progress
INFO [03-31|14:18:43.365] Imported new chain segment
INFO [03-31|14:18:43.365] Syncing: chain download in progress
INFO [03-31|14:18:43.365] Syncing: state download in progress
WARN [03-31|14:18:43.365] Synchronisation failed, retrying
WARN [03-31|14:18:43.366] Unexpected account range packet
INFO [03-31|14:18:43.373] Looking for peers
WARN [03-31|14:18:45.003] Snap syncing, discarded propagated block
INFO [03-31|14:18:53.368] Syncing: state download in progress
INFO [03-31|14:18:53.370] Syncing: chain download in progress
INFO [03-31|14:18:53.371] Syncing: state download in progress
WARN [03-31|14:18:53.374] Unexpected account range packet
INFO [03-31|14:18:53.374] Syncing: state download in progress
INFO [03-31|14:18:53.374] Syncing: state healing in progress
INFO [03-31|14:18:53.385] Rebuilding state snapshot
INFO [03-31|14:18:53.385] Committed new head block
INFO [03-31|14:18:53.385] Resuming state snapshot generation
INFO [03-31|14:18:53.386] Generated state snapshot
INFO [03-31|14:18:53.385] Imported new chain segment
INFO [03-31|14:18:53.392] Looking for peers

```

16. Now we will interact with the blockchain using javascript console

geth attach node1/geth.ipc

```

pranav@DESKTOP-OSHSLJ1:~/private_ethereum_setup$ geth attach node2/geth.ipc
Welcome to the Geth JavaScript console!

instance: Geth/v1.12.0-stable-e501b3b0/linux-amd64/go1.20.3
at block: 32 (Tue Mar 31 2026 14:20:10 GMT+0000 (UTC))
datadir: /home/pranav/private_ethereum_setup/node2
modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> net.peerCount
1
>

```

17. Interacting with the blockchain

